Example 1 - rivt Doc

Attn: **user example**

project: **0001**

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1. Summary

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual.

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command of the Doc API (rv.D) at the end of the file. Published files are found in the sub-folders of the _published folder.

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2. Load Combinations

Table 1: ASCE 7-05 Load Effects (saved as csv in _stored folder)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

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3. Loads and Geometry

Successive lines of value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 lb_in2	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the src folder. The description is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines).

Table 3: Beam Geometry from file: **src/beam1.csv**

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing

variable	value	[value]	description
spn_1	16.00 ft	4.88 m	beam span

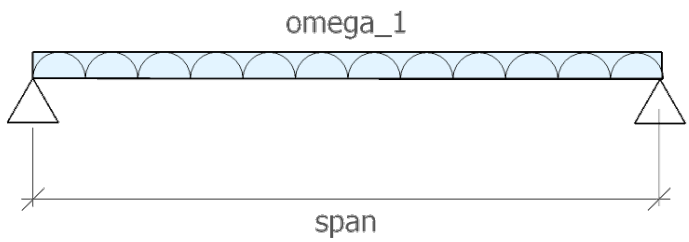

Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: dead load [ASCE7-05 2.3.2]

```
dl₁ = 1.2·(D₄ + spc₁·(D₁ + D₂ + D₃))

dl₁ = 3.64 klf     [dl₁] = 53.09 kN_m   | dead load [ASCE7-05 2.3.2]

D₁          D₃              spc₁          D₂          D₄
_________   _____________   ____________  __________  __________________
3.80 psf    10.00 psf       2.00 ft       2.10 psf    3.00 klf
joists DL   partitions DL   beam spacing  plywood DL  fixed machinery DL
_________   _____________   ____________  __________  __________________
```

Eq. 2: live load [ASCE7-05 2.3.2]

```
ll₁ = 1.6·L₁·spc₁

ll₁ = 0.13 klf     [ll₁] = 1.87 kN_m   | live load [ASCE7-05 2.3.2]

L₁           spc₁
___________  ____________
40.00 psf    2.00 ft
ASCE7-05 LL  beam spacing
___________  ____________
```

Eq. 3: total load [ASCE7-05 2.3.2]

```
ω₁ = dl₁ + ll₁

ω₁ = 3.77 klf     [ω₁] = 54.96 kN_m   | total load [ASCE7-05 2.3.2]
```

```
ll₁                          dl₁
──────────────────────────   ──────────────────────────
128.00 ft·psf                3.64 klf
live load [ASCE7-05 2.3.2]   dead load [ASCE7-05 2.3.2]
──────────────────────────   ──────────────────────────
```

---

4. Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions from file: **src/sectprop.py**

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S

```
section₁ = rectsect(b₁, h₁)

section_1  = 540.00 in3     [section_1 ] = 8849.01 cm3   | rectangle - S

b_1         h_1
──────────  ──────────
10.00 inch  18.00 inch
──────────  ──────────
```

Eq. 5: rectangle - I

```
inertia₁ = rectinertia(b₁, h₁)

inertia_1  = 4860.0 in4     [inertia_1 ] = 202288.5 cm4   | rectangle - I

b_1        h_1
─────────  ─────────
10.0 inch  18.0 inch
─────────  ─────────
```

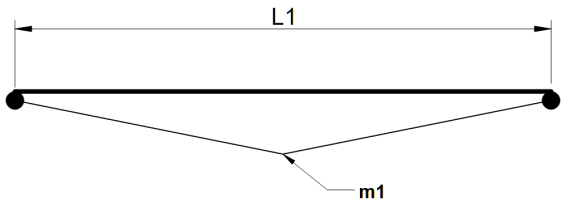

Fig. 2 - Moment Diagram

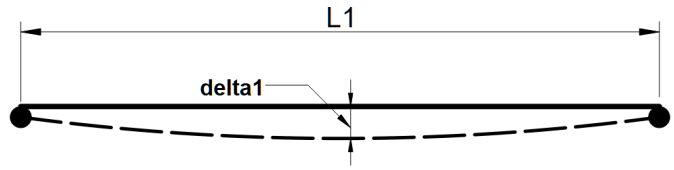

Fig. 3 - Deflection Diagram

Eq.6

$$\sigma_1 = \frac{M_1}{S_1}$$

Eq. 7: mid-span UDL moment

$$m_1 = \frac{\omega_1 \cdot spn_1^2}{8}$$

m_1 = 120.52 ftkip [m_1] = 163.40 mkN | mid-span UDL moment

spn_1	ω_1
16.00 ft beam span	3.77 klf total load [ASCE7-05 2.3.2]

Eq. 8: bending Stress

$$fb_1 = \frac{m_1}{section_1}$$

fb_1 = 2678.2 lb_in2 [fb_1] = 18.5 MPA | bending Stress

m_1	$section_1$
120.5 ft2·klf	540.0 inch3

```
mid-span UDL moment  rectangle - S
___________________  __________
```

Eq. 9: stress ratio

```
fb₁ < Fb₁                                                                  ▮
                                                                           ▮
fb₁       Fb₁        ratio     check    reference                         ▮
_______   _________  _______   ______   ___________                       ▮
2.68 ksi  20.00 ksi  0.133908  OK       stress ratio                      ▮
_______   _________  _______   ______   ___________                       ▮
```

Eq. 10: mid-span deflection

```
δ₁ = midspan_δ(spn₁, ω₁, E₁, inertia₁)

delta_1  = 0.04 inch     [delta_1 ] = 1.00 mm   | mid-span deflection

spn_1     omega_1    inertia_1      E_1
_______   _________  ____________   ___________
16.00 ft  3.77 klf   4860.00 inch4  29000.00 ksi
_______   _________  ____________   ___________
```